

EE 210L: Digital Circuits and Logic Design Lab

Lab 1: Logic Gates and Truth Tables

Name: _____ Date: _____

Partner(s): _____ Section: _____

Fill in every measured value. Record the voltage you actually measured, not the value you expected. Show your work where a calculation is asked for.

Part 5.1: Power Supply Setup

Measured power supply voltage: _____ V

Current limit setting: _____ mA

Part 5.2: First Circuit, AND Gate (74LS08)

A	B	Measured V_{out} (V)	Logic Output	Expected Output
0	0			
0	1			
1	0			
1	1			

Part 5.3: OR, NOT, NAND, NOR, XOR

For each gate, set the inputs with jumper wires, measure the output with the DMM, and convert the measurement to a logic level ($HIGH \geq 2.0 V$, $LOW \leq 0.8 V$).

OR (74LS32)

A	B	Measured V_{out} (V)	Logic Output	Expected Output
0	0			
0	1			
1	0			
1	1			

NOT (74LS04)

A	Measured V_{out} (V)	Logic Output	Expected Output
0			
1			

NAND (74LS00)

A	B	Measured V_{out} (V)	Logic Output	Expected Output
0	0			
0	1			
1	0			
1	1			

NOR (74LS02)

Remember that the 74LS02 output is pin 1 and its inputs are pins 2 and 3.

A	B	Measured V_{out} (V)	Logic Output	Expected Output
0	0			
0	1			
1	0			
1	1			

XOR (74LS86)

A	B	Measured V_{out} (V)	Logic Output	Expected Output
0	0			
0	1			
1	0			
1	1			

Did every gate match the truth table you predicted in your pre-lab? If any did not, explain what was wrong and how you fixed it.

Part 5.4: Undefined Region

With one input of the 74LS08 tied HIGH and the other left disconnected:

Voltage measured at the floating input pin: _____ V

Voltage measured at the output: _____ V

Which band of §3.1 does the floating input fall into (valid HIGH, valid LOW, or undefined)? Why should you never leave an input floating in a real design?

Part 5.5: Propagation of a Real Signal

Input at pin 1	V at pin 1 (V)	V at pin 2 (V)	V at pin 4 (V)
Logic 0			
Logic 1			

Explain what the second inverter does to the signal level coming out of the first one.

Part 5.6: NAND as a Universal Gate

5.6.1 NOT from NAND

Draw your circuit diagram in the box. Label every pin number you used.



Figure 1: NOT gate built from a single NAND gate.

A	Measured V_{out} (V)	Logic Output
0		
1		

5.6.2 AND from NAND

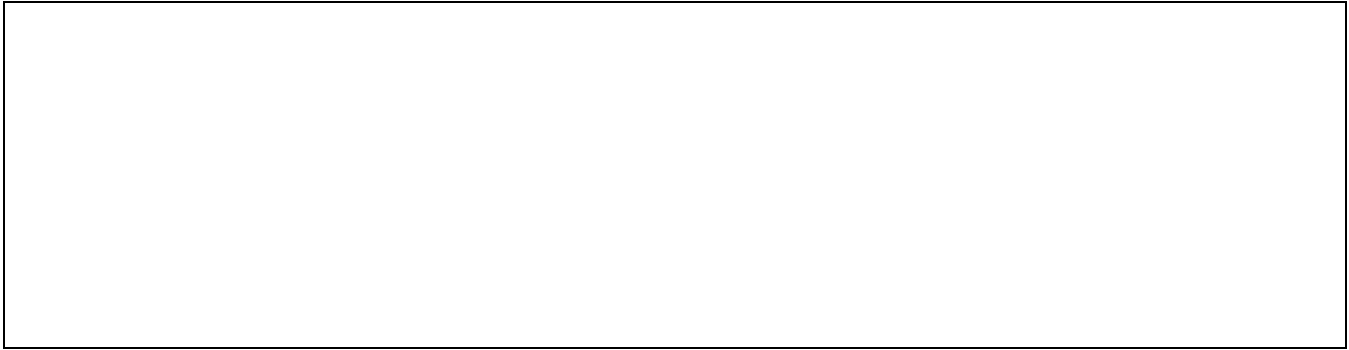


Figure 2: AND gate built from NAND gates only.

A	B	Measured V_{out} (V)	Logic Output
0	0		
0	1		
1	0		
1	1		

5.6.3 OR from NAND

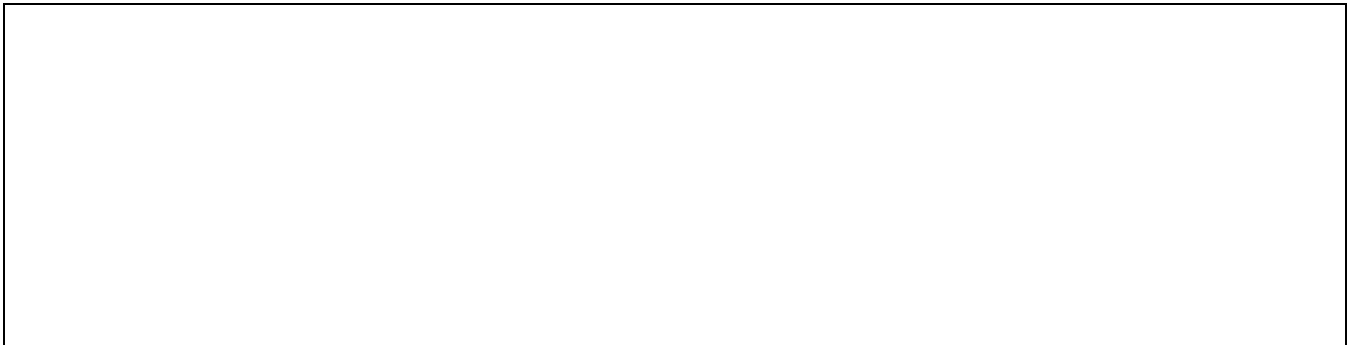


Figure 3: OR gate built from NAND gates only.

A	B	Measured V_{out} (V)	Logic Output
0	0		
0	1		
1	0		
1	1		

How many NAND gates did each circuit take? Why does being able to build NOT, AND, and OR out of one gate type matter to someone manufacturing a chip?

Discussion and Conclusions

Summarize what you measured and what it confirmed. Address at least: how your measured output voltages compared to an ideal 0 V / 5 V, any wiring mistakes you made and how you found them, and what you would check first if a gate gave you the wrong output.
